



ENT-164 · Intro to Making

BEGINNER'S GUIDE · MACOS · NO EXPERIENCE NEEDED

Making with AI

OpenCode + DeepSeek V4.1 Flash

Download the OpenCode app, connect it to OpenRouter, and start asking questions. No terminal, no coding — just a normal Mac app.

About 15 minutes

For macOS

Provider: OpenRouter

Model: DeepSeek V4.1 Flash



The OpenCode start screen: the opencode logo, the message box, and DeepSeek V4.1 Flash already selected.

OpenCode is an AI helper that runs as an app on your Mac. You type to it in plain English, and it can explain code, help with class projects, and walk you through technical tasks. **DeepSeek V4.1 Flash** is the AI "brain" that writes the answers, and **OpenRouter** is the service that connects the app to it.

Think of it this way: OpenCode is your phone, OpenRouter is your phone company, and DeepSeek V4.1 Flash is the person you're calling. This guide sets all three up. You won't need to use a terminal or write any code.

Why DeepSeek V4.1 Flash for this class?

- **Built for this work** — it's designed for reasoning and agentic coding with reliable tool use, so it can genuinely help with your class projects.
- **Open weights** — the model itself is openly licensed and inspectable, which fits a making class better than a closed black box.
- **Big context, tiny cost** — it can read entire files (up to 1M tokens) for a fraction of a cent per question, so our shared class budget lasts the term.
- **One connection** — because OpenRouter gives us access to it, you set up one account and one key, and you can explore other models later without starting over.

What you'll need

- ✓ An Apple laptop running macOS 13 or newer
- ✓ An internet connection
- ✓ Your Tufts email — you'll sign up with it in Step 3 to join the class workspace
- ✓ Nothing to pay — the class budget covers AI usage for your coursework

New word alert. **OpenRouter** is a website that gives one account access to many AI models from different companies. Instead of signing up for each company separately, you connect OpenRouter once and pick any model you like.

1 Download the OpenCode app

1. Open your browser and go to **opencode.ai/download**.
2. Click the big **Download for macOS** button. The site automatically picks the right version for your Mac.
3. The download starts — you'll find the file in your **Downloads** folder.

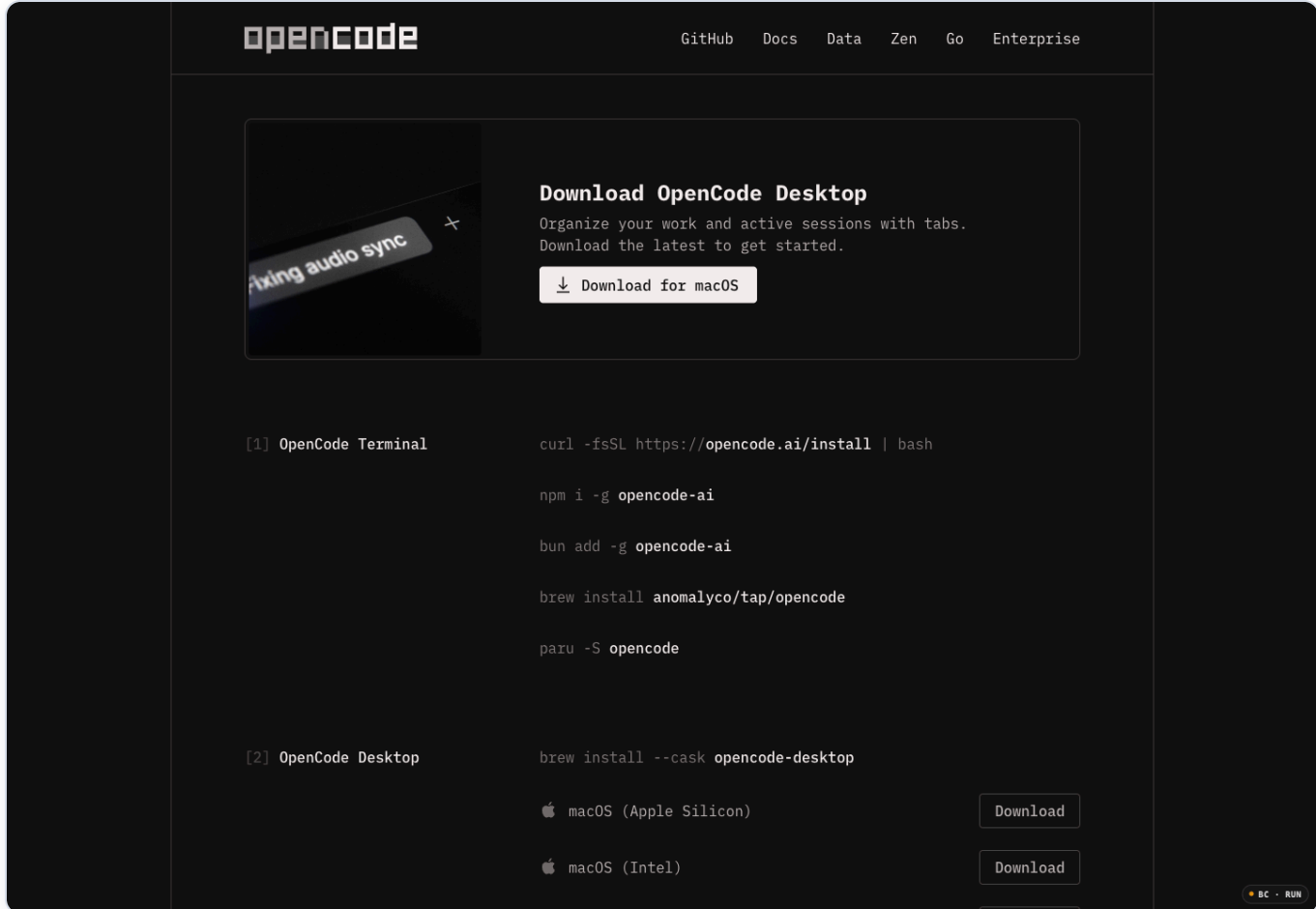


Figure 1. The download page. Use the download button in the banner at the top — it detects whether your Mac needs the Apple Silicon or Intel version.

2 Install the app

1. Open the downloaded file (click it in your **Downloads** folder, or in the notification Safari shows).
2. A small window opens showing the **OpenCode** icon and your **Applications** folder.
3. **Drag the OpenCode icon onto the Applications folder**, exactly like the arrow shows.
4. Close that window, then open **OpenCode** from your Applications folder (or press `⌘ + Space`, type **OpenCode**, and press `Enter`).

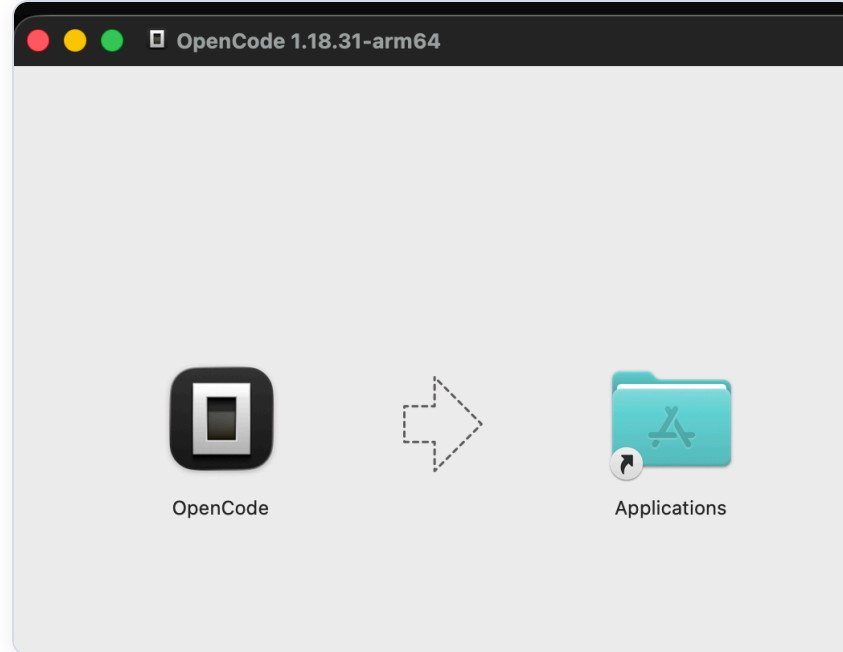


Figure 2. Installing is just one drag: move the OpenCode icon onto the Applications folder.

macOS says it "cannot verify" the app? That's the standard warning for apps downloaded from the internet. Open your **Applications** folder, right-click (or Control-click) **OpenCode**, choose **Open**, and confirm. You only need to do this once. The app is signed and comes directly from `opencode.ai`.

3 Join your class on OpenRouter

Your class has its own workspace on OpenRouter. You join it with your **Tufts email** — no invitation email needed.

1. Go to **openrouter.ai** and click **Sign Up**.
2. Click the **Google** button and sign up with your **Tufts email** (for example `yourname@tufts.edu`). Using Google skips creating a separate password.
3. OpenRouter suggests your class workspace — click **Request to join**.
4. Your instructor approves the request. The class workspace (for example **ENT-164 Intro to Making**) then appears in the workspace menu at the top-left of the page.

Use your Tufts email. The class workspace is tied to the `tufts.edu` domain. If you sign up with a different email, you won't be offered the workspace or access to the class budget.

4 Create your own API key

An **API key** is a long password that lets the app use OpenRouter on your behalf. Yours will start with `sk-` or `-`. Treat it like a bank password: don't share it or post it anywhere.

1. In the top-left of the page, make sure your class workspace (**ENT-164 Intro to Making**) is selected — your key belongs to the workspace that's active when you create it.
2. Open your account menu and click **Keys**.

3. Click **Create API Key**, name it **OpenCode**, and copy the key — the site shows it only once. Paste it somewhere temporary (Notes is fine) for the next step.
4. If the workspace shows a credit balance, you're ready to go — and **you don't need to add money or a card**. The class budget covers your usage.

Your access is on the class budget. The instructor pays for AI usage through a shared class budget, so there's nothing for you to buy. Because that budget is shared, please keep your questions focused on class work and avoid long, off-topic chats — that keeps it available for everyone through the term.

5 Connect OpenRouter in the app

1. Open OpenCode. At the bottom you'll see the message box and, below it, a button with the current model's name (it starts as **Big Pickle**).
2. Click that model button. A panel opens listing models and providers.
3. Scroll to **Add more models from popular providers** and click **See 70+ more providers**.
4. Find **OpenRouter** (it's in the **Popular** section) and click it.
5. Paste your sk-or-... key into the box and click **Continue**.

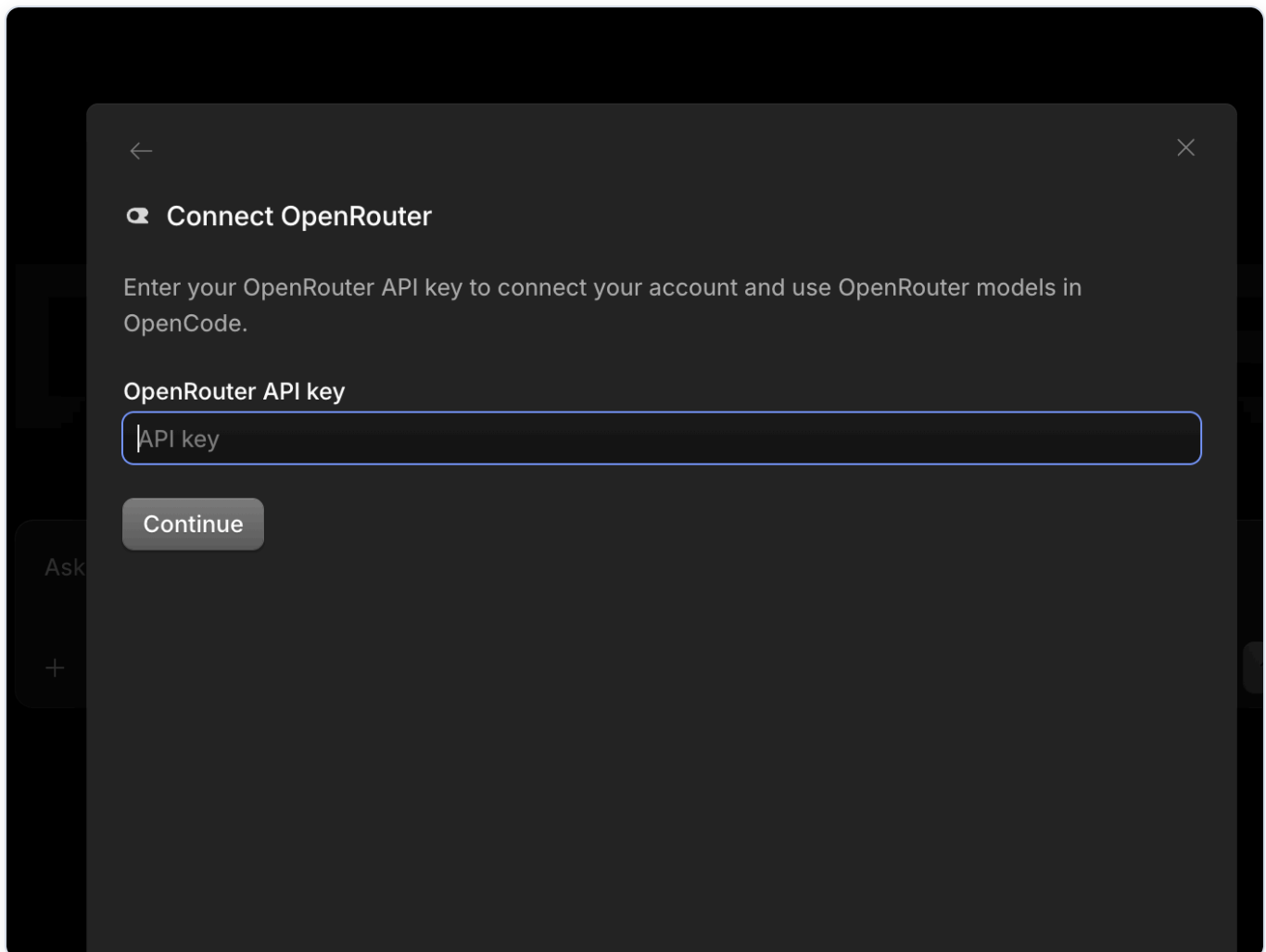


Figure 3. The Connect OpenRouter dialog. Paste the key you copied in Step 4 and click **Continue**.

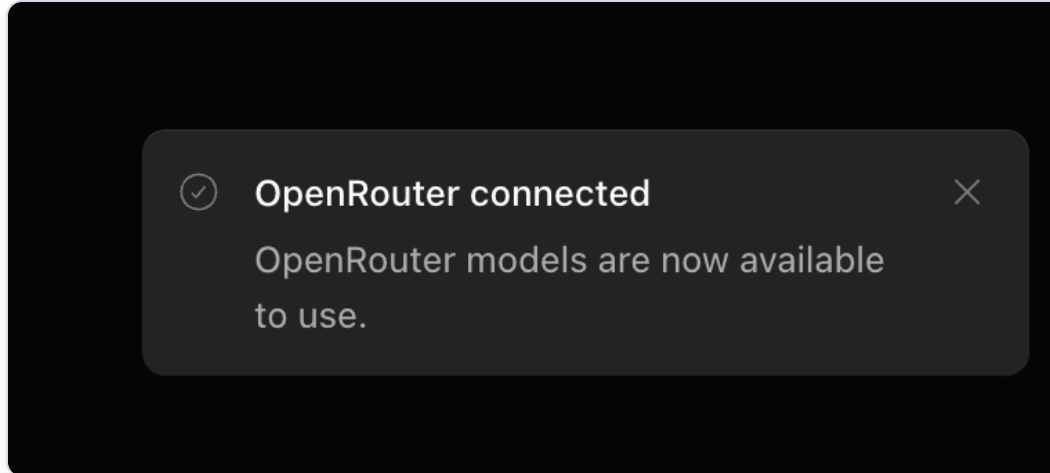


Figure 4. Success — a message in the corner confirms **OpenRouter connected**. You only do this once; the app remembers the key.

6 Choose DeepSeek V4.1 Flash

Click the model button under the message box again, then:

1. Type **v4.1 flash** in the search box at the top of the panel.
2. If **DeepSeek V4.1 Flash** appears, click it and you're done with this step.
3. If it doesn't appear yet, click **Manage models** at the bottom, type **v4.1 flash** in its search box, and turn on the switch next to **DeepSeek V4.1 Flash**.
4. Close that panel, search again in the model list, and click **DeepSeek V4.1 Flash**.

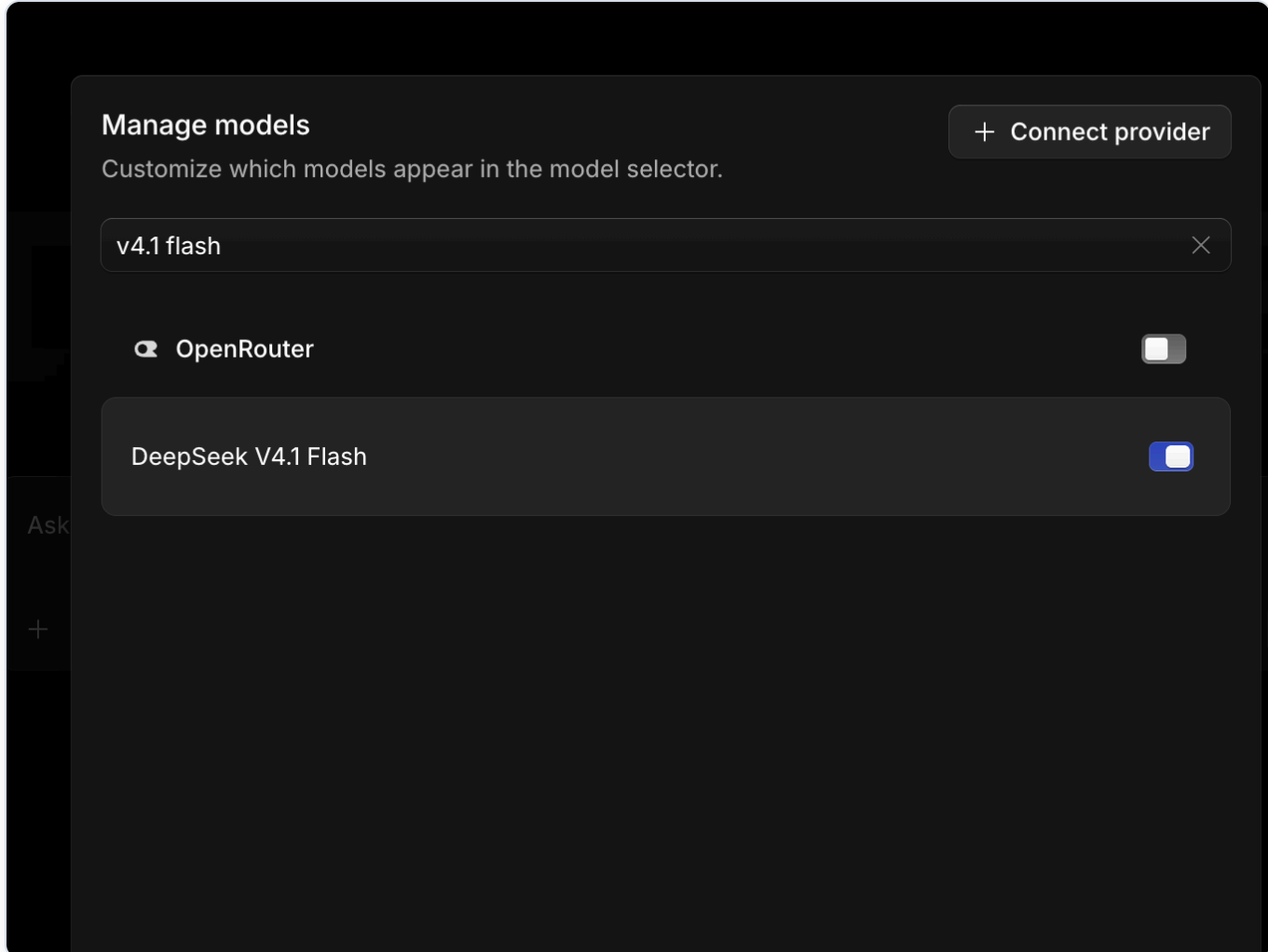


Figure 5. In **Manage models**, search for the model and switch it on so it appears in the model list.

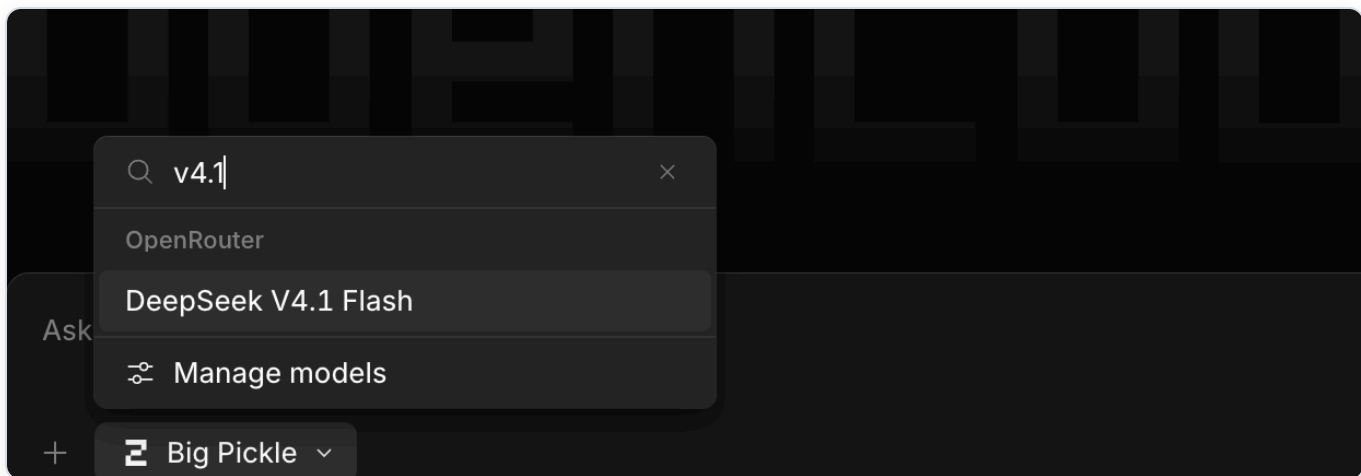


Figure 6. Back in the model list: type **v4.1** and click **DeepSeek V4.1 Flash** (listed under OpenRouter).

Did it work? The button under the message box now shows **DeepSeek V4.1 Flash**. That's the model every new question will use.

7

Ask your first question

1. Click the message box at the bottom that says "Ask anything..."
2. Type a question in plain English, for example: *"I'm new to coding. What can you help me with?"*

3. Press **Enter** (or click the arrow button at the right) and wait a few seconds.

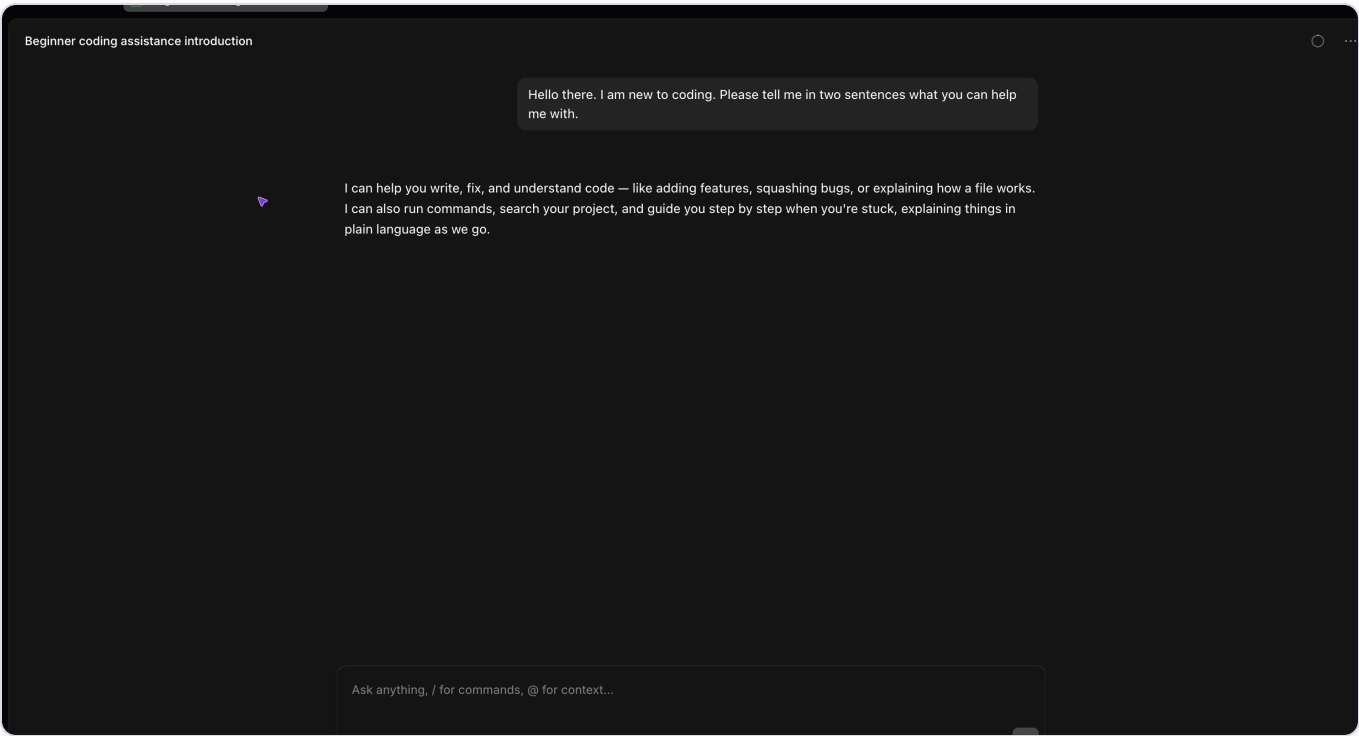


Figure 7. Your question appears in the bubble at the top; the answer gets written below it. The model's name is always visible under the message box.

To quit: press **⌘ + Q** like any Mac app. To come back later, just open OpenCode again — your conversations are saved as tabs at the top.

If something goes wrong

WHAT YOU SEE	WHAT TO DO
macOS won't open the app	In Applications, right-click (Control-click) OpenCode and choose Open . If needed: System Settings → Privacy & Security → Open Anyway .
Can't find the provider panel	Click the button showing the current model name, just below the message box. Everything is reached from there.
OpenRouter isn't listed	Make sure you clicked See 70+ more providers — OpenRouter is in the Popular section at the top.
"Invalid API key"	The key was copied incompletely. Create a new key at openrouter.ai (Keys) and paste it again, starting with sk-or- .
An error about credits or balance	Don't add credit yourself. Check that the class workspace is selected at the top-left, then tell your instructor — the shared class budget may be out of funds.
DeepSeek V4.1 Flash isn't in the list	Use Manage models in the model panel, search v4.1 flash , and switch it on.

WHAT YOU SEE	WHAT TO DO
No answer, or it seems stuck	Wait ten seconds — the first reply can be slow. Check your internet, then ask again.  +  and reopening also helps.

Words you'll see

OpenCode	The app on your Mac where you chat with the AI.
OpenRouter	The service that connects the app to many AI models. Needs an account and an API key.
API key	A secret password (starts with sk-or–) that lets the app use your OpenRouter account. Never share it.
Provider	The company or service that runs the AI connection — here, OpenRouter.
Model	The specific AI "brain" you talk to — here, DeepSeek V4.1 Flash.
Session / tab	One conversation. Each tab at the top of the app is a separate chat.

Where everything lives

WHAT YOU WANT	WHERE TO CLICK
Join the class workspace	Sign up at openrouter.ai with your Tufts email, then Request to join
Create your API key	openrouter.ai → account menu → Keys → Create API Key
Start a new conversation	The + next to the tabs at the very top
Change the model	The model button under the message box (shows "DeepSeek V4.1 Flash")
Connect another provider or fix your key	Model button → See 70+ more providers
Show more models in the list	Model button → Manage models
Quit the app	 + 

Once you're comfortable, the official docs at opencode.ai/docs show what else OpenCode can do. There's no rush — asking questions is a great way to learn.